Water and agriculture:

evidence from subsistence farmers in Peru*

Fernando M. Aragón† Mahyar Rezazadeh‡

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Abstract

This paper examines the impact of access to water sources (such as rainfall or irrigation water) on agricultural outcomes in the context of subsistence farmers. Using microdata from Peruvian households and a novel re-analysis dataset on precipitation, we document a positive effect of rainfall and agricultural output. This effect is driven mainly by increments in input use. We also document impacts on crop mix and child labor. Furthermore, we find suggestive evidence that formal, secure, access to other water sources, such as irrigation water, has beneficial effects on agricultural production.

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†Department of Economics, Simon Fraser University; email: faragons@sfu.ca

‡Department of Economics, Simon Fraser University; email: smr20@sfu.ca

1 Introduction

Several studies document a positive relationship between access to water and agricultural production. Through its impact on agriculture, water can affect other important outcomes such as income, conflict, health and human capital accumulation.[1]

However, most of the evidence focuses on rainfall shocks using data from the Indian subcontinent and sub-Saharan Africa. In places with prevalent rainfed agriculture, rainfall is indeed a good measure of access to water. However, in settings with limited rainfall, access to other water sources, such as reservoirs or surface irrigation water, could be more important, and thus current estimates may not be information about the relationship between water and agriculture. This is an important limitation, given that this relationship is a key element in understanding the impact of shocks on water security, which may occur due to competition for water sources from urban growth or climate change.

This paper contributes to the debate by examining the impact of rainfall shocks and access to irrigation water on agricultural outcomes in Peru. Our empirical analysis combines survey microdata from farming households with weather data from satellite imagery and re-analysis datasets. The identification strategy takes advantage of the variation in weather within the locality. We start by estimating the relation between rainfall shocks on agricultural output and crop yields. Then we extend the analysis to study how farmers respond productively, like changing land and labor use, including child labor and crop mix. Finally, we examine how access to other water sources, such as irrigation, affects agricultural output.

We find evidence of a positive relationship between rainfall shocks and agricultural output. The magnitude is economically relevant and comparable to previous estimates from India and Brazil. The effect is greater on the coast than in the highlands and appears to be driven by positive, rather than negative, rainfall shocks. This effect on output appears to be

[1]See, for instance, Hoddinott and Kinsey (2001), Miguel et al. (2004) Maccini and Yang (2009), Shah and Steinberg (2017). For a more comprehensive review of the literature, see Dell et al. (2014).

entirely driven by increases in land and labor use. This result is consistent with farmers facing binding water constraints. Therefore, when there is greater availability of water (owing to more rain), farmers can use more of their land and expand their agricultural activities.

We also found evidence that positive rainfall shocks affect crop mix (increasing cash crops such as rice and maize and reducing the area planted with tubers) and child labor. However, there is significant regional variation: the relationship between rainfall and child labor is negative in the highlands but insignificant on the coast.

Finally, we examine the role of security of access to surface waters. This is relevant since, in the Peruvian context, a substantial proportion of farmers rely on surface waters for irrigation, especially on the semi-arid coast. Water users can obtain licenses that provide formal rights of use and priority access to water sources. We find that changes in river discharge are positively associated with agricultural output, but only in areas where most farmers have water licenses. This finding highlights the importance of water rights in shaping farmers' ability to benefit from existing water sources.

2 Subsistence farming in Peru

Our empirical analysis focuses on subsistence farmers in rural Peru. This sector has low productivity and shares many characteristics with subsistence agriculture in other developing countries: it is mainly composed of small productive units (i.e., households), with low capital intensity and low levels of technology adoption.

Table 1 presents some key summary statistics of farmers in our sample.[2]. Most farmers are poor and depend on agriculture as their main source of income. The incidence of poverty in our sample of farmers is around 40%. The average farm is around 2 hectares, has a low degree of specialization, and uses practices akin to traditional rather than industrial farming.

[2]Data sources and variable definitions are described in Section 3.1

They rely on domestic labor (including child labor), cultivate a variety of crops instead of monocropping, and leave some land uncultivated. Some of this uncultivated land is reported to be fallow while the rest is covered with grasses, bushes, and forests. These last uses are also consistent with fallowing and crop rotation, but we cannot rule out that part of this land is nonagricultural.

Table 1: Summary statistics

	All	Coast	Highlands
	(1)	(2)	(3)
A. Household characteristics			
HoH age	54.8	57.4	54.2
HoH is female	18.7	11.4	20.1
Household size	4.3	4.6	4.2
Poor (%)	40.7	24.2	43.8
HoH complete primary educ. (%)	53.7	59.4	52.6
Child works (%)	15.7	7.7	17.3
B. Agricultural characteristics			
Value of agricultural output	1152.5	3619.8	681.9
Area planted (has)	2.1	2.5	2.0
No. HH members work in farm	2.3	2.2	2.3
Hire workers (%)	45.0	54.7	43.1
Use fertilizers (%)	78.4	77.7	78.6
Use pesticides (%)	50.3	66.9	47.1
Uncultivated land (% land holding)	43.7	15.4	49.1
Irrigated land (% land holding)	38.9	85.4	30.1
No access to irrigation (%)	37.4	6.1	43.4
Tubers (% total output)	31.7	6.9	36.4
Cereals (% total output)	30.9	40.0	29.2
C. Environmental conditions during last growing season			
Average temperature (℃)	24.9	35.8	22.8
Harmful Degree Days (HDD)	2.5	5.6	1.9
Monthly rainfall (mm)	101.3	18.2	117.2
River discharge (m^3/s)	239.9	44.7	277.2
No. obs	24,802	4,007	20,795

Notes: HoH= household head. HH= household. Output in 2007 USD.

We focus on two climatic regions: the coast and the highlands.[3] These two regions exhibit different climates due to their proximity to the sea and altitude. The coast is a narrow strip that extends from the seashore up to 500 meters above sea level (masl). It has a semi-arid climate, with warm temperatures and very little precipitation. The highlands extend from 500 masl up to almost 7,000 masl, although most agriculture stops below 4000 masl. It has a much cooler and wetter climate, with seasonal precipitations in spring and early summer.[4]

These climatic differences are associated with different agricultural practices: coastal farmers are more dependent on irrigation, while agriculture in the highlands is a combination of irrigated and rainfed agriculture. The main source of irrigation water is rivers and lakes and to a lesser extent reservoirs. In both areas, there is very limited groundwater use. Coastal farmers are also less likely to be poor and to have a different crop mix, cultivating a greater share of fruits and cereals. Similar to other subsistence farmers in the developing world, farmers in this study represent a large fraction of the rural poor, have relatively small farms, and practice traditional agricultural techniques.

Planting occurs throughout the year (see Figure A.3 in the online appendix). However, it is concentrated between October and March. These months correspond to spring and summer in the southern hemisphere and are considered the main growing season in Peru. In the empirical analysis, we use this period as the relevant time frame of exposure to link agricultural outcomes to temperature and precipitation.

[3]Peru has three main climatic regions: the coast to the west, the Andean highlands, and the Amazon jungle to the east. We do not include observations from the jungle due to the poor quality of the temperature and precipitation data.

[4]See Figures A.1 and A.2 in the online appendix for the distribution of precipitation and temperature during the growing season in both regions.

3 Methods

3.1 Data

We combine household surveys with satellite imagery and re-analysis data to construct a comprehensive dataset containing agricultural, socioeconomic, and environmental variables. We restrict the sample to households with agricultural activities located on the coast and highlands. The final data set consists of around 65,400 household-year observations spanning years 2007 to 2016. Table 1 presents the summary statistics for the sample.

Household and agricultural data The main data source is repeated cross sections of the Peruvian Living Standards Survey, an annual household survey collected by the National Statistics Office (ENAHO). This survey is collected on a continuous and rolling basis.

To measure real agricultural production, we construct a Laspeyres index using the quantity produced from each crop and baseline local prices.[5] We calculate land used by adding the size of parcels dedicated to seasonal and permanent crops. We distinguish between domestic and hired labor. We measure hired labor using the self-reported wage bill paid to external workers in the last 12 months. To measure domestic labor, we use the information on household members' employment. In particular, we calculate the number of household members working in agriculture and build an indicator of child labor.[6]

The survey also provides information on sociodemographic characteristics, agricultural practices, and farm conditions (such as intercropping, access to irrigation and use of fertilizers), and the geographical coordinates of each primary sampling unit or survey block.[7] In rural areas, this corresponds to a village or cluster of dwellings. We use this geographical

[5] As local prices, we use the median price of each crop in a given department (n=24) in 2007.

[6] Child labor is defined as an indicator equal to one if a child living in the household aged 6-14 reports doing any activity to obtain some income. This includes helping on the family farm, selling services or goods, or helping relatives, but excludes household chores.

[7] There are around 3,800 unique coordinate points in our sample.

information to link the household data to satellite imagery and re-analysis data.

We complement the household survey with soil quality data from the Harmonized World Soil Database (FAO and ISRIC, 2012) and district-level data from the 2012 Peruvian Agricultural Census on access to irrigation and tenure of water licenses.

Precipitation, river discharge, and temperature We use the SENAMHI PISCOp data set developed by Aybar et al. (2020). This data set is a gridded re-analysis product that combines data from monitoring station data and satellite imagery (CHIRPS) Beck et al. (2017). We use version 1.2. This version contains data on average monthly rainfall during the period 1981-2016 with a resolution of 0.1 degrees.[8]

In several contexts, especially on the semi-arid coast, irrigation is the main source of water. We proxy the availability of surface irrigation water using monthly river discharge measures (i.e., the amount of water flowing in a river channel) collected by SENAMHI.[9] An important limitation of this dataset is its sparse geographical coverage. There are only 28 monitoring stations with historical records that cover the period of analysis. This feature means that even with a broad definition of a river basin, there are several parts of the country with missing data on river discharge.[10]

Finally, we complement the rainfall data with information on local temperature. We use satellite imagery from the MOD11C1 product provided by NASA (Wan et al., 2015). This product provides daily measures of daytime temperature on a grid of 0.05×0.05 degrees, equivalent to 5.6 km squares on the equator, and is already cleaned of low-quality readings

[8]There is also an updated version that includes years 2017-2020. We prefer not to use this latest version in the baseline analysis since we understand that it is still unstable. However, we present some preliminary analysis using this updated version (see Column 6 in Table 3).

[9]Ideally, we would like to combine rainfall data with information on the amount of water available to or used by the irrigation system (for example, usable volume in the reservoir or lake, or amount of water collected in the irrigation intake). However, this information is not available.

[10]Dividing the country in 59 river basins, there are only monitoring stations in 16 of them (see Figure A.4 in the online appendix).

and processed for consistency.[11]

To link the weather and household data, we attribute to a given household the weather conditions in the cell that overlap its coordinates (or river basin in the case of discharge data). Then, we aggregate weather data to obtain measures of exposure to weather during a specific growing season (i.e., months October to March).

In the case of precipitation and river discharge, we construct a standardized index measuring deviations from historical means in terms of standard deviations. In particular, we defined the index as $\frac{X_{it}-\bar{X}_i}{\sigma_{X_i}}$ where X_{it} is rainfall (or river discharge) during the growing season t in location i, $\bar{X}_i$ is the historical average and σ_{X_i} is the standard deviation. Note that this measure is similar to the Standardized Precipitation Index (SPI), a widely-used measure of rainfall shocks.

For temperature, we follow the methodology used by Aragón et al. (2021) and construct measures of degree days(DD) and harmful degree days (HDD) during the growing season. DD measures cumulative exposure to temperatures between a lower bound, usually 8℃ up to an upper threshold τ, while HDD captures exposure to temperatures above τ. The inclusion of HDD allows for potentially different, non-linear effects of extreme heat.[12]

Figures 1 and 2 display a map of Peru with the average precipitation and temperature (at the district level) during the growing season on the coast and highlands. These figures corroborate the initial observations that the coastal areas near the Pacific Ocean are hotter and drier.[13]

[11]The satellite data provides estimates of land surface temperature (LST) not of surface air temperature, which is the variable measured by monitoring stations. For that reason, the reader should be careful when comparing the results of this paper to other studies using re-analysis data or station readings. LST is usually higher than air temperature, and this difference tends to increase with the roughness of the terrain. However, both indicators are highly correlated (Mutiibwa et al., 2015).

[12]Following Aragón et al. (2021), we use a value of $\tau = 33$℃ for both climatic regions. The main results are robust to the use of a wide range of τ values, including region-specific thresholds.

[13]See Figures A.2 and A.1 in the online appendix for the distribution of rainfall and temperature in both climatic regions.

Figure 1: Average monthly rainfall during growing season

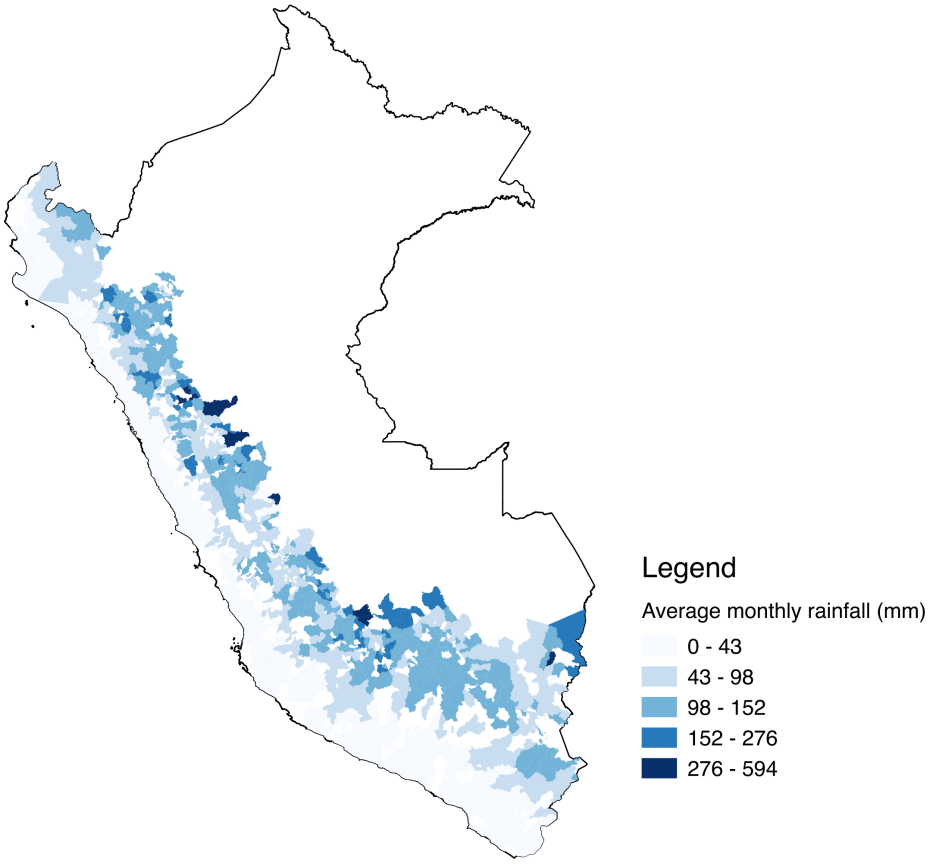
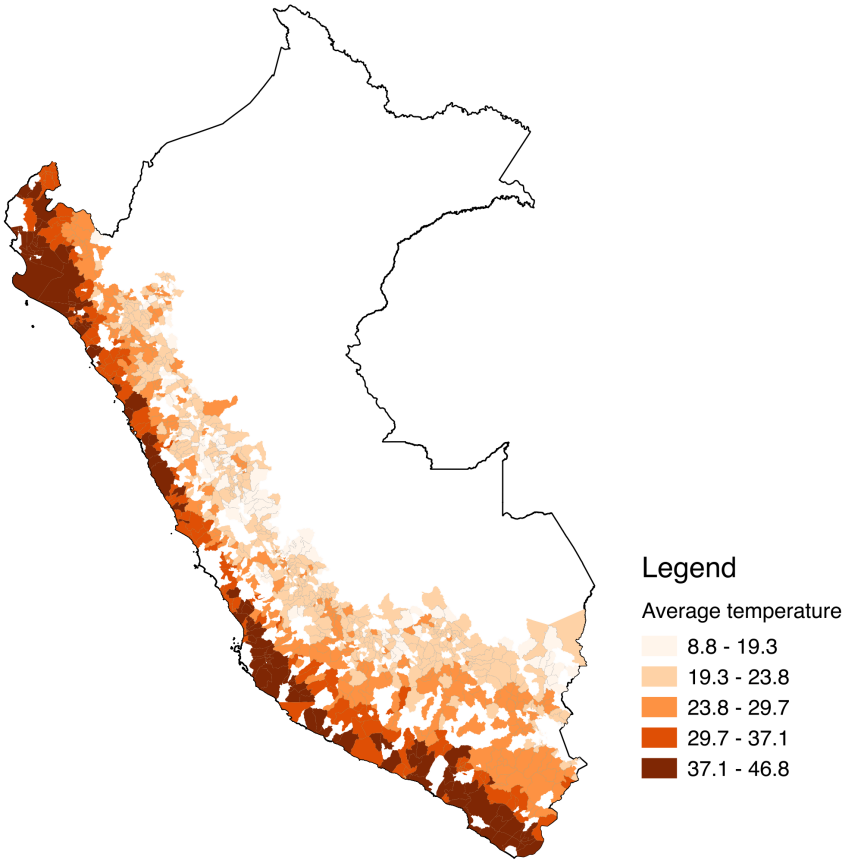


Figure 2: Average temperature during growing season



3.2 Empirical strategy

The empirical analysis aims to study how water supply shocks (such as changes in rainfall) affect agricultural output and farmer's input choices. To do so, we estimate the following reduced-form baseline regression:

$$\ln y_{ijt} = g(\gamma, \omega_{it}) + \phi X_i + \rho_{jt} + \epsilon_{ijt}, \tag{1}$$

where the unit of observation is farmer i in location (region or basin) j and growing season t. y is our dependent variable (such as agricultural output or measures of input use) of input use and $g(\gamma, \omega_{it})$ is a linear combination of measures of rainfall and temperature (DD and HDD). The baseline specification does not include measures of river discharge due to the substantial amount of missing data and the loss of observations. Therefore, the use of river discharge may lead to an increase in measurement error and loss of precision. However, since river discharge may be a relevant omitted variable, we also check the robustness of the main results to include this additional covariate.

The baseline specification also includes a vector of farmer characteristics,X_i, such as age, sex, and level of education, and climatic region-by-growing season and region-by-month of interview fixed effects (ρ_{jt}). These fixed effects account for the initial and time-varying regional heterogeneity, as well as for region-specific seasonality. We also check the robustness of the results using a richer set of fixed effects and control variables (see column 1 in Table 3). We cluster the errors at the district level ($n = 1,133$) to account for potential geographical and serial correlation.

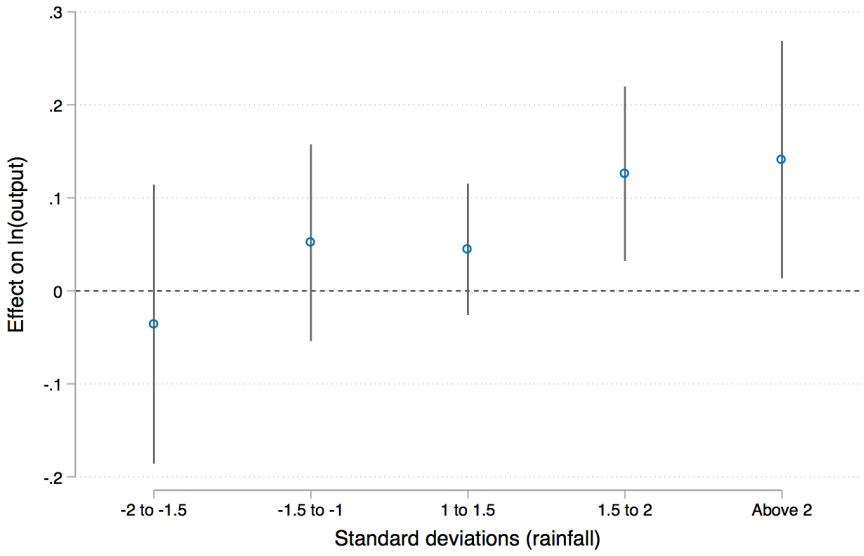
4 Results

This section presents the main empirical results on the effect of rainfall shocks on agricultural outcomes. We start by documenting the relationship between rainfall and agricultural output. Then examine the productive responses of the farmers. Finally, we examine the role of water security in shaping the effects of irrigation water.

4.1 Effect of rainfall on agricultural

Figure 3 depicts our main finding. It shows the estimated effect of rainfall shocks (in terms of standard deviations from the historical mean) on agricultural output. We observe a positive relationship, especially when rainfall shocks are positive: above-mean levels of rainfall are associated with greater output, while below-mean precipitation is associated with a negative (albeit statistically insignificant) lower output.

Figure 3: Effect of rainfall on agricultural output



Notes: Figure displays estimates of regressing ln(output) on bins of rainfall deviations from historical mean. Circles represent point estimates, lines display 95% confidence interval. The omitted category is rainfall within ±1 standard deviations from the mean.

We examine this finding in more detail by estimating the baseline specification (1). The results, displayed in Table 2, suggest a positive and statistically significant relationship between rainfall and agricultural production: estimates imply that one standard deviation increase in rainfall is associated with an increase of almost 4% of agricultural output (column 1). Columns 2 and 3 divide the sample by climatic region.[14] In both cases, the estimates are positive albeit the magnitude seems greater on the coast.

These findings are consistent with several studies documenting a positive relationship between rainfall and rural income, especially in settings with rainfed agriculture [15]. For example, in the case of Brazil, Hidalgo et al. (2010, figure 2) find that, at low to moderately high levels, there is a positive relationship between rain and agricultural income. Their estimates suggest that one standard increase in rainfall increases income by around 4%. However, this relationship becomes negative at high precipitation levels (above 2 standard deviations). Similarly, in the case of India, Sarsons (2015, table 2) documents a positive relation between rainfall and agricultural output. In particular, levels of rainfall above the 80th percentile increase output by 0.054, while levels below the 20th percentile decrease output by a similar magnitude.[16]

Columns 4 to 6 check the robustness of the main results, adding measures of river discharge. As discussed above, we use this variable as a proxy for the availability of irrigation water. This variable could be a relevant explanatory variable, and thus its omission would create endogeneity concerns. The pattern of results remains similar, although the point estimates of the effect of rainfall are higher than in columns 1-3. Interestingly, the estimates of river discharge (around -0.043 for the whole sample) are weakly negative, which is consistent

[14] In addition to differences by climatic region, we also examine potential heterogeneous effects by gender, education, poverty status, and presence of children. In all these cases, the differences are not statistically significant (see Table B.1 in the online appendix).

[15] See for instance Miguel et al. (2004), Sarsons (2015) and references in Dell et al. (2014).

[16] We replicate this last specification using the same definition of rain shock and obtain an estimate equal to 0.051 (see Table B.2 in the online appendix).

with damages due to flooding and waterlogging.

However, these results have an important limitation: more than half of the observations are lost due to missing data on river discharge. This feature creates concerns about possible measurement error and selection bias. For this reason, we suggest that the reader interpret these last results with caution.

Table 2: Rainfall and agricultural output

Dep. Variable:	ln(output)			ln(output)		
	All	Coast	Highlands	All	Coast	Highlands
	(1)	(2)	(3)	(4)	(5)	(6)
Standardized rainfall	0.038**	0.124*	0.028*	0.098***	0.105	0.075**
	(0.016)	(0.066)	(0.016)	(0.029)	(0.073)	(0.031)
Standardized river discharge				-0.043*	0.013	-0.052**
				(0.025)	(0.080)	(0.025)
Basin FE	No	No	No	Yes	Yes	Yes
Temp. controls	Yes	Yes	Yes	Yes	Yes	Yes
No. obs.	65,419	9,367	56,052	27,517	4,839	22,678
R-squared	0.129	0.069	0.090	0.163	0.121	0.124

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics: household head education, age and gender. Columns 4 to 6 also include basin fixed effects (n= 15).

Alternative measures of rainfall shocks Our baseline results use standardized rainfall (i.e., deviations from the historical mean in terms of standard deviation) as the preferred measure of rainfall shock. We check the robustness of the main results using alternative measures. In particular, we replicate columns 1 to 3 of Table 2 using two alternative measures of rainfall shock. First, we use logarithmic deviations from the historical mean $lnX_{it} - ln\bar{X}_i$, where X_{it} is rainfall during the growing season and $\bar{X}_i$ is the historical average. This measure has been used in some studies on the effect of rainfall on income such as Björkman-Nyqvist

(2013). Second, we use a categorical variable equal to 1 if rain levels are above the location's 80th percentile, -1 if they are below the 20th percentile, and 0 otherwise. This measure of rainfall shock has been used by Sarsons (2015) in their study of rainfall on agricultural output and violence in India, and is similar to the one used by Jayachandran (2006) and Kaur (2019).

The results, shown in TableB.2 in the online appendix, suggest that the main results of a positive effect of rainfall on output are robust to using these alternative measures, although they become less precise for the smaller subsamples.

Additional checks Table 3 presents additional checks. Column 1 adds a richer set of controls including flexible trends by department (n=24), month of interview-by-year fixed effects, and indicators of soil quality. Column 2 trims the top and bottom 1% of the sample based on farm size. In both cases, the results remain similar to the baseline specification.

Columns 3 and 4 examine the effects of using data from the Amazon jungle. The main results remain robust to adding the jungle observations to the baseline sample (column 3). However, when the sample is restricted to the jungle only (column 4), it becomes evident that the effect of rainfall on output in this region is statistically insignificant, and the point estimate is even negative. An interpretation is that changes in rainfall do not affect agricultural production. This is possible given the already high levels of precipitation in the jungle. However, an alternative explanation is that the null results are reflecting attenuation bias due to measurement error. As acknowledged by the developers of the PISCOp dataset, the data in the Amazon region is of lower quality (Huerta et al., 2018). This lower quality is due to a combination of fewer monitoring stations and poorer satellite data associated with cloud and vegetation coverage.

Measurement error could also explain the statistically insignificant results estimated in columns 5. Column 5 expands the PISCOp dataset to cover the years 2017 to 2019. This

additional data were, at the time of preparing this paper, still under revision and therefore susceptible to contain measurement errors.

Column 6 estimates the baseline specification using MERRA2 as an alternative data source of rainfall. MERRA2 is a re-analyzed gridded dataset that combines satellite and rain gauge data, providing information on daily and monthly precipitation across the globe. The results of column 6 suggest that the primary findings regarding the positive impact of rainfall on output are robust to using this alternative dataset.

Table 3: Additional checks

Dep. Variable:	ln(output)					
	(1)	(2)	(3)	(4)	(5)	(6)
Standardized rainfall	0.046**	0.030*	0.022*	-0.022	0.015	0.026**
	(0.020)	(0.017)	(0.013)	(0.021)	(0.013)	(0.011)
Robustness check	Adding more controls	Trimmed sample	Adding jungle	Jungle only	Adding years 2017-19	Using MEERA2 data
Observations	65,418	52,234	88,861	23,442	81,983	60,597
R-squared	0.193	0.137	0.175	0.065	0.127	0.128

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) and farmer characteristics as the baseline specification in column 1 of Table 2. Column 1 adds department-by-growing season and month of interview-by growing season fixed effects plus indicators of soil quality. Column 2 trims the top and bottom 1% of the sample based on farm size. Column 3 and 4 use data from the Amazon jungle. Column 5 expands the original dataset with PISCOp data for years 2017-2019, while column 6 uses measures of standardized precipitation based on MERRA2 re-analysis data instead of PISCOp.

Why would rainfall increase output? Consider an agricultural production function $y = f(A, land, labor, water)$ where A is total factor productivity. In this simple framework, an increase in rainfall can affect agricultural output by at least two channels. The first channel is through changes in productivity A. This would occur, for example, by reducing damages associated with extreme heat or by allowing cultivation of higher-value, but more

water-demanding, crops. A second channel is through, affecting the total amount of inputs. By increasing the amount of water, rainfall would have a positive effect on output by itself. Furthermore, to the extent that water complements other inputs, it could also lead to increases in land and labor use.

To distinguish between these channels, we examine the effect of rainfall on measures of land productivity (yields) and on output conditional on input use (land and labor). As measures of land and labor, we include the logarithm of the planted area, the wage bill of hired workers, and the number of household members working on the farm.

The results, shown in Table 4, suggest that there is no significant effect on land productivity (columns 1 to 3). They also show that the impact of rainfall on output disappears, or become only marginally significant, when controlling for input use (columns 4 to 6). These results suggest that most of the effect of rainfall is driven by changes in land and labor, not by changes in residual productivity or water itself. One interpretation of this finding is that farmers face water constraints that limit their ability to use their available land and labor. This assumption is reasonable in the context of semi-arid conditions and rainfed agriculture found on the Peruvian coast and highlands.

4.2 Effect of rainfall on other outcomes

Land and labor The previous results suggest possible effects of rainfall on input use. In particular, if water is complementary to other inputs (such as land and labor), then the increase in water would imply a weak increment in input use. We examine this hypothesis by estimating reduced-form unconditional factor demands. In particular, we modify the baseline regression (equation (1)) by using a measure of input use (such as the log of the planted area or indicators of pesticide use) as a dependent variable and including, as additional controls, proxies of household labor and land endowments (i.e., log household size and area owned).

The results, displayed in Table 5, suggest that rainfall leads to an increase in the use of

Table 4: Rainfall, land productivity and residual output

Dep. Variable:	ln(output per ha)			ln(output)		
	All	Coast	Highlands	All	Coast	Highlands
	(1)	(2)	(3)	(4)	(5)	(6)
Standardized rainfall	-0.008	0.046	-0.012	-0.008	0.074*	-0.013
	(0.020)	(0.050)	(0.021)	(0.013)	(0.041)	(0.013)
ln(area planted)				0.283***	0.437***	0.255***
				(0.013)	(0.026)	(0.013)
ln(wage bill)				0.189***	0.238***	0.170***
				(0.005)	(0.011)	(0.005)
No. HH members work in farm				0.140***	0.059***	0.162***
				(0.008)	(0.022)	(0.007)
No. obs.	65,260	9,343	55,917	27,451	4,819	22,632
R-squared	0.049	0.070	0.024	0.128	0.133	0.108

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specification in column 1 of Table 2. Columns 4 to 6 also include measures of agricultural inputs (land and labor).

several inputs. The effect on land, in particular, is of comparable magnitude to the effect of rainfall on output. We also observe positive estimates on measures of hired labor, both in terms of the likelihood of hiring workers (column 2) and the amount spent on hired workers' wages (column 3). A more nuanced picture emerges with other inputs such as pesticides and fertilizers. While there is a positive relationship between rainfall and pesticide use, the relationship with fertilizer use is negative (columns 4 and 5). This last result could reflect a certain degree of substitutability between fertilizer and water supply.

Columns 6 to 8 examine changes in how land is used. In particular, we examine possible changes in crop mix. These changes could occur, for example, if crops have different water demands or if the additional cultivated land is suitable for different crops. To study the effects on crop mix, we group crops into two categories: tubers (mostly potatoes) and other

crops. Tubers are the most important crop among Peruvian subsistence farmers and account for almost 30% of the value of agricultural output and 15% of the planted area. We find that additional rainfall reduces the area cultivated with tubers but increases the area planted with other crops (columns 6 and 7). This change in crop mix results in the reduction of tubers as a proportion of total output (column 7).

Table 5: Rainfall and input use

Dep. Variable:	Land	Labor		Other inputs		Crop Mix		
						ln(output)		Tubers
	ln(area planted)	Hire workers	ln(wage bill)	Use pesticides	Use fertilizers	Tubers	Other crops	% output
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standardized rainfall	0.036**	0.031***	0.159***	0.014***	-0.017***	-0.211***	0.106***	-0.028***
	(0.018)	(0.005)	(0.029)	(0.005)	(0.005)	(0.027)	(0.023)	(0.004)
Endow. Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Temp. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	65,260	65,419	65,417	65,419	65,419	65,419	65,419	65,419
R-squared	0.258	0.047	0.082	0.055	0.030	0.287	0.194	0.227

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specification in column 1 of Table 2. They also control for household endowments using log of household size and log of area of land owned.

Child labor and education The observed effects of rainfall on output and input use have implications for other relevant outcomes, such as child labor and education. According to the model presented in Section 2.2, an increase in household income (due to a greater agricultural output) may have a weakly negative effect on child labor. The increased usage of external workers as a complementary workforce for child labor increases the substitution of external workers with children. As a result of these two forces, child labor is expected to decrease as a result of a rainfall shock. In turn, these changes in child labor could have repercussions on schooling decisions.

We examine this issue in Table 6. We restrict the sample to households with at least one child between the ages of 6 and 15 and examine two outcomes: an indicator of child work (columns 1 to 3) and the proportion of children enrolled in school (columns 4 to 6).[17]

The results paint a mixed picture. On average, rainfall appears to reduce the likelihood of child labor (column 1). This last result is consistent with findings in the literature documenting a positive correlation between income shocks and child labor (Bandara et al., 2015, Beegle et al., 2006). However, this estimate hides regional heterogeneity: the relation is positive for the coast but negative for the highlands (columns 2 and 3).

The results on school enrollment suggest no significant effect. However, these findings should be interpreted with caution. They are at best a preliminary exploration of potential effects of rainfall on education but, due to data limitations, provide only a partial picture. A more comprehensive study would need to examine other relevant outcomes such as school attendance, test performance, or repetition grades. Similarly, the ENAHO survey relies on self-declared enrollment, which may be subject to systematic measurement error.

[17]Child labor is defined as an indicator equal to one if a child living in the household aged 6-15 reports doing any activity to obtain some income. The list of activities considered work includes helping on the family farm, selling services or goods, or helping relatives, but excludes household chores.

Table 6: Rainfall, child labor and enrollment

Dep. Variable:	Child works			% children enrolled in school		
	All	Coast	Highlands	All	Coast	Highlands
	(1)	(2)	(3)	(4)	(5)	(6)
Standardized rainfall	-0.020**	-0.005	-0.0203*	0.055	0.553	0.111
	(0.010)	(0.019)	(0.0108)	(0.471)	(0.612)	(0.501)
Endow. Controls	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. Var.	0.307	0.208	0.320	77.4	75.4	77.6
No. obs.	8,791	960	7,831	8,791	960	7,831
R-squared	0.16	0.18	0.154	0.700	0.757	0.702

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specification in column 1 of Table 2. They also control for household endowments using log of household size and log of area of land owned. Sample is restricted to agricultural households with children 6 to 15 years old.

4.3 Surface water and water security

So far, the analysis has focused on rainfall as a source of water for agricultural production; however, irrigation water is also important in several contexts. For example, around 50% of the farmers in the highlands use irrigation water. This proportion is much higher, near 95%, in the semi-arid coastal valleys (see Table B.4 in the online appendix). The main sources of irrigation water are rivers, lakes and reservoirs (only 2.6% of farmers use groundwater).

Importantly, access to irrigation water is not guaranteed, but is shaped by different forms of water rights. In the Peruvian context, water rights are granted through licenses, permits, and authorizations. Licenses grant permanent, guaranteed access to a water source. Licenses usually specify the maximum amount of water that the licensee is able to collect, subject to water availability.

On the contrary, permits grant access only to residual or surplus water available *after* license holders have used their allocation. Authorizations, on the other hand, provide short-term (usually 2 years) access to a water source to carry out specific activities such

as soil remediation, infrastructure development, and hydrological studies. These rights can be granted to individuals or legal entities, such as firms and water users' organizations, but they are not transferable. Moreover, in practice, there is a high degree of informality, with many farmers without a license or permit. According to the 2012 Agricultural Census, only a quarter of farmers using irrigation waters have a license.

This institutional setup creates different levels of water security: likely higher for license holders, but lower for permit holders or informal water users. To examine the role of water security and irrigation water, we calculate the proportion of irrigation users (at the district level) who hold a water license. Then we estimate the model in columns 4-6 in Table 2, adding an interaction of this institutional measure with the standardized river discharge. The key idea is to examine whether the physical availability of surface water has a greater effect in areas where farmers have priority and guaranteed access to it.

Table 7 presents the results. There are two relevant observations. First, the effect of rainfall on agricultural output remains positive and significant. Second, the effect of river discharge (a proxy of available surface water) is increasing in the prevalence of water licenses and becomes positive in districts when more than half of irrigation users have a water license (see Figure A.5 in the online appendix).

These findings highlight the importance of water security. One possible interpretation is that more secure access to irrigation water (owing to the tenure of a water license) allows farmers to better use the available water resources. Consistent with this interpretation, we find that the effects are greater on the coast, a region in which agriculture is more dependent on irrigation, than in the highlands (columns 2 and 3). Similarly, we observe that river discharge only seems to matter in districts where the main source of irrigation water is a river, but becomes insignificant in areas where lakes or reservoirs are more important (columns 4 and 5).

However, these findings are preliminary and are intended to guide future research in

this area. As acknowledged above, there are substantial data limitations when using the data on river discharge: it fails to include other relevant water sources (such as lakes and reservoirs) and it has sparse geographical coverage. Furthermore, the measure of the water allocation institution (that is, the prevalence of water licenses) is potentially endogenous and may pick up other confounding factors, such as the quality of irrigation infrastructure or organizational capacity. Addressing these concerns to get a better understanding of the role of water security is, however, beyond the scope of this paper.

Table 7: River discharge, output and water rights

Dep. Variable:	ln(output)				
				Main irrigation source	
	All	Coast	Highlands	River	Reservoir, lake or groundwater
	(1)	(2)	(3)	(4)	(5)
Standardized rainfall	0.104***	0.129*	0.075**	0.133***	0.059*
	(0.029)	(0.076)	(0.031)	(0.044)	(0.034)
Standardized river discharge	-0.095***	-0.190**	-0.071**	-0.123***	-0.040
	(0.030)	(0.093)	(0.029)	(0.041)	(0.036)
Stand. river discharge × % has water license	0.212***	0.407***	0.092	0.234***	0.107
	(0.070)	(0.119)	(0.077)	(0.085)	(0.089)
% has water license	0.247	0.526	0.188	0.295	0.193
No. obs.	27,467	4,821	22,646	14,503	12,964
R-squared	0.164	0.140	0.125	0.181	0.205

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specification in columns 4-5 of Table 2. *% has water license* adopts values between 0 and 1.

5 Conclusion

This paper studies the effect of rainfall on agricultural outcomes in the context of Peruvian subsistence farmers. Consistent with findings in other contexts, we document a positive effect of rainfall on agricultural output. This result is mostly driven by an increment in land and labor use. We also document changes in crop mix and child labor, and present suggestive evidence of the importance of water security in the context of irrigated farms.

Understanding how rainfall and, more broadly, water availability affect farmers is relevant due to expected changes in precipitation and drought patterns due to climate change and increasing pressures from other water users. This paper contributes to this debate by providing estimates for a new context and examining in more detail farmers' productive responses.

Interestingly, our findings point out the importance of institutions (such as water rights) in shaping how the availability of irrigation water affects agriculture. We are unable, however, to explore this issue in more detail due to data limitations and lack of quasi-experimental variation in the measure of water rights. However, these issues warrant further research.

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ONLINE APPENDIX
– Not for publication–

A Additional Figures

Figure A.1: Distribution of average temperature during growing season, by climatic region

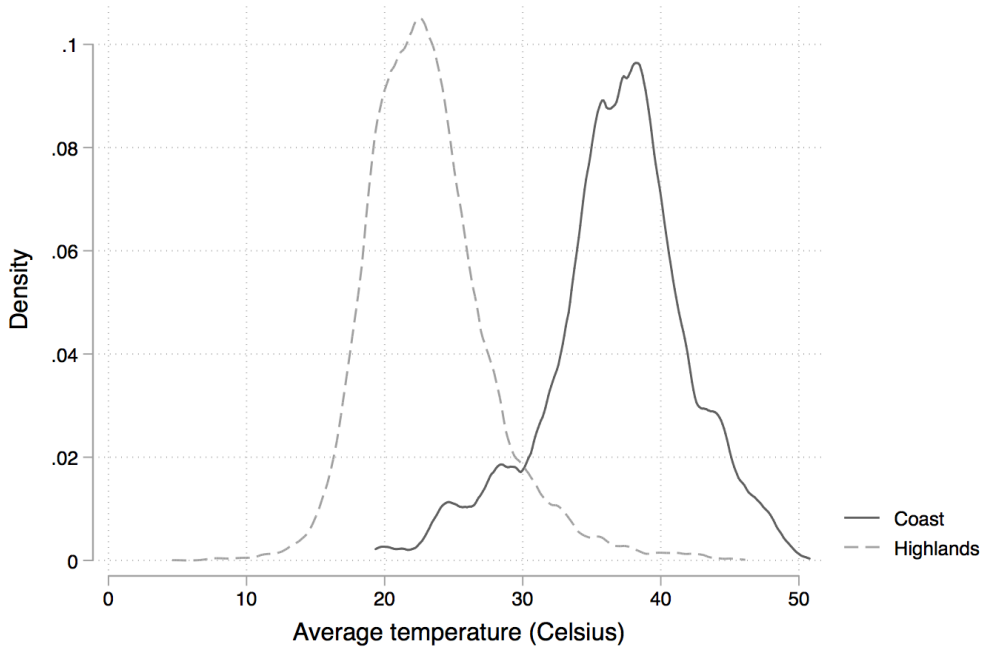


Figure A.2: Distribution of average rainfall during growing season, by climatic region

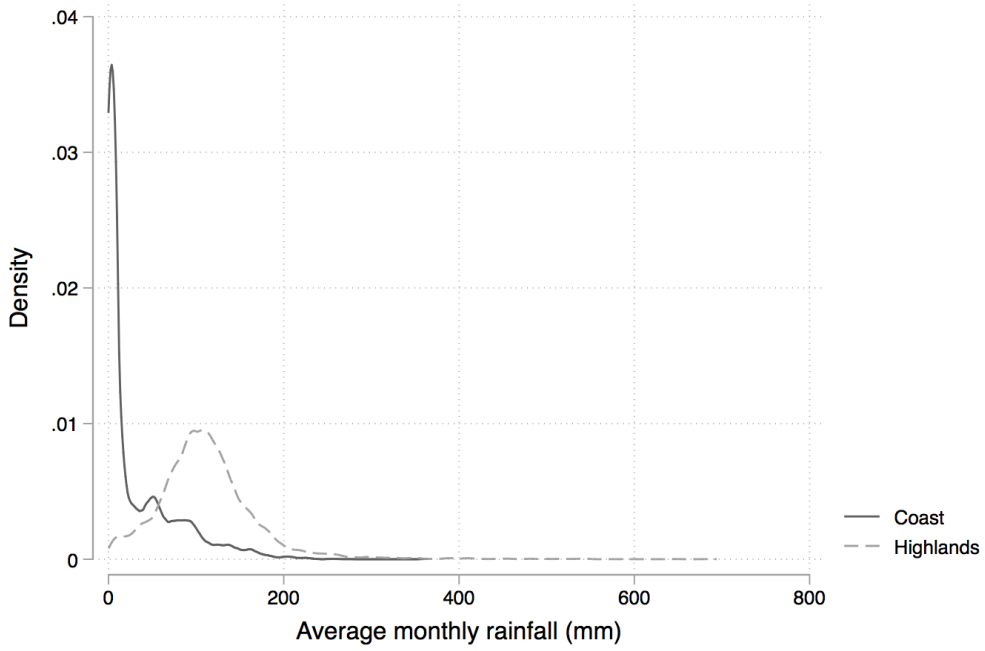
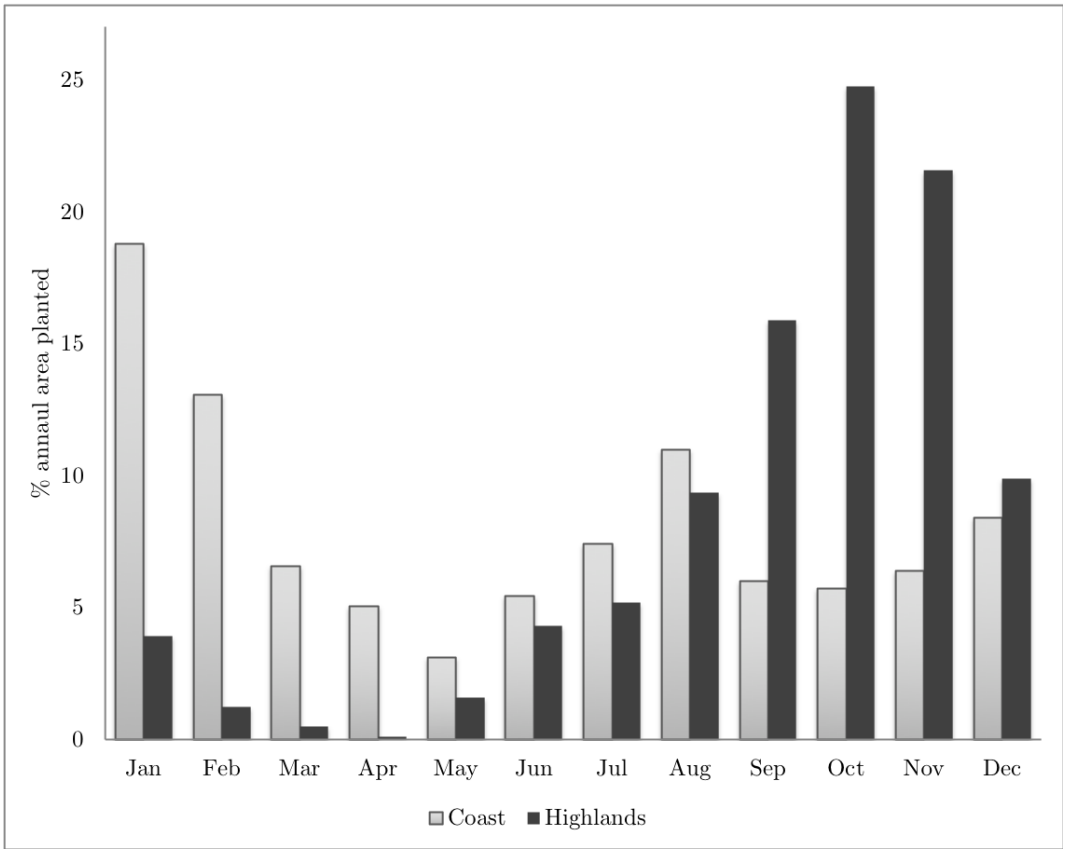


Figure A.3: % of annual area planted in a month, by climatic region



Source: National Agricultural Survey of Peru, 2014-2017.

Figure A.4: River basins and discharge monitoring stations

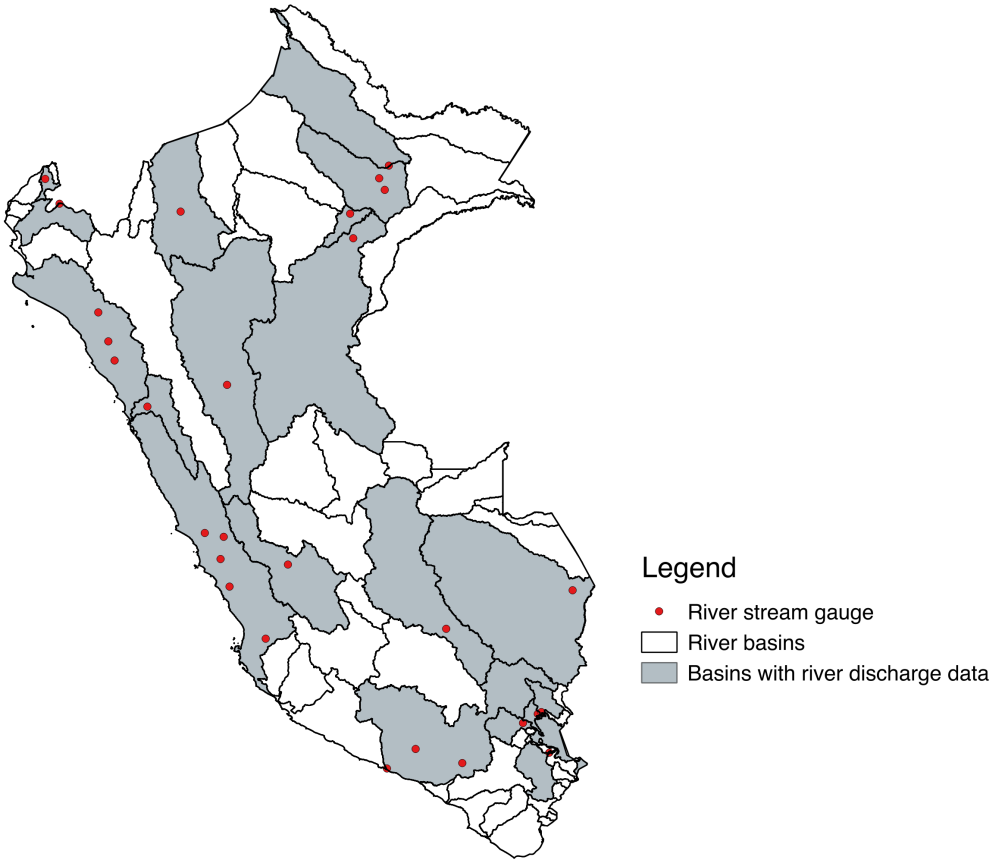
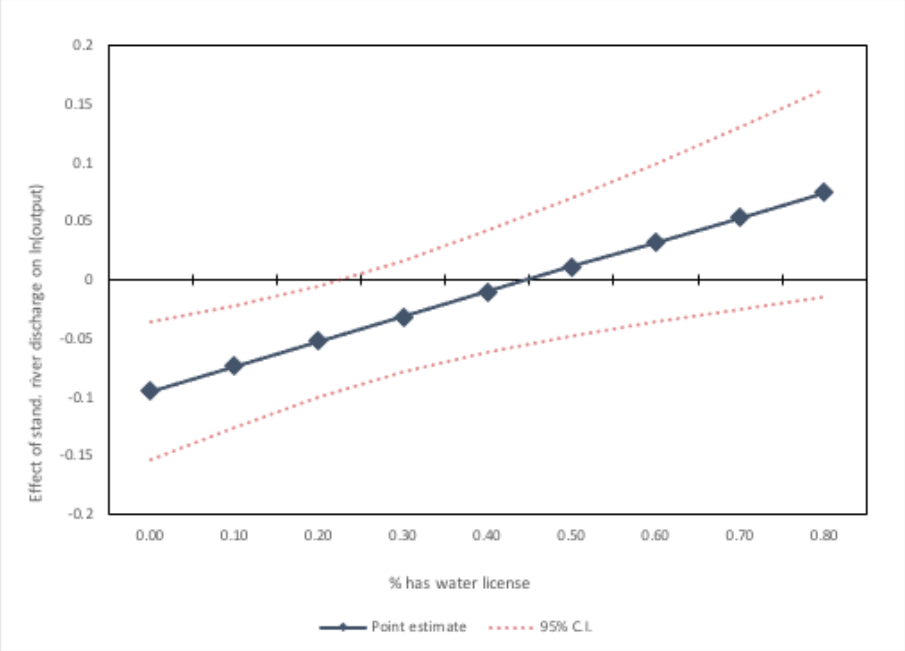


Figure A.5: Effect of river discharge on output, by % has water license



Notes: Figure display the estimated effect of standardized river discharge on ln(output) at different values of $ has water license using results in column 1 in Table7. Circles represent point estimates, lines display 95% confidence interval.

B Additional tables

Table B.1: Exploring heterogeneous effects

Dep. Variable:	ln(output)				
	(1)	(2)	(3)	(4)	(5)
(A) Standardized rainfall $\times$ $X = 0$	0.042** (0.017)	0.042** (0.017)	0.036* (0.021)	0.040** (0.018)	0.046** (0.018)
(B) Standardized rainfall $\times$ $X = 1$	0.020 (0.022)	0.036* (0.019)	0.035** (0.015)	0.022 (0.019)	0.030* (0.018)
p-value (A)-(B)	0.222	0.720	0.967	0.386	0.274
$X = 1$ if	female	Completed primary educ.	is poor	is extreme poor	has child 6-15 years old
No. obs.	65,419	65,419	65,419	65,419	65,419
R-squared	0.129	0.130	0.143	0.138	0.130

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specifications in Table 2. Regression include interactions of the measure of rainfall with and a binary variable X that indicates whether the household head is female, completed primary education, poverty status of the household, and presence of a child between 5-14 years old.

Table B.2: Replication of Table 2 using alternative measures of rainfall shock

Dep. Variable:	ln(output)			ln(output)		
	All	Coast	Highlands	All	Coast	Highlands
	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall shock (log dev. from mean)	0.089*	0.135**	-0.007			
	(0.046)	(0.059)	(0.056)			
Rainfall shock à la Sarsons (2015)				0.051**	0.116	0.042*
				(0.023)	(0.086)	(0.023)
Basin FE	No	No	No	No	No	No
Temp. controls	Yes	Yes	Yes	Yes	Yes	Yes
No. obs.	65,410	9,358	56,052	27,508	4,830	22,678
R-squared	0.129	0.070	0.090	0.163	0.123	0.123

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specifications in Table 2. Columns 4 to 6 use as a measure of rainfall shock a categorical variable equal to 1 if rain levels in the growing season are above the 80th percentile, -1 if they are below the 20th percentile and 0 otherwise,.

Table B.3: Rainfall, heterogeneous child labor and enrollment

Dep. Variable:	Child works			% children enrolled in school		
	All	Coast	Highlands	All	Coast	Highlands
Standardized rainfall	-0.0220*	-0.00559	-0.0232*	-0.0949	0.608	-0.0551
	(-2.10)	(-0.28)	(-2.07)	(-0.20)	(0.60)	(-0.11)
Female × Standardized rainfall	0.0247	0.00132	0.0243	1.284	-0.685	1.386
	(1.43)	(0.03)	(1.35)	(1.34)	(-0.34)	(1.38)
Female	0.00300	-0.0710	0.00629	-1.722	-2.232	-1.586
	(3.86)	(2.83)	(3.90)	(13.78)	(5.45)	(12.51)
No. obs.	8791	960	7831	8791	960	7831
R-squared	0.16	0.18	0.154	0.70	0.76	0.70

Notes: Standard errors (in parenthesis) are clustered at the district level. All specifications include climatic region-by-growing season and climatic region-by-month of interview fixed effects, temperature controls (DD and HDD) as well as farmer characteristics as the baseline specification in column 1 of Table 2. They also control for household endowments using log of household size and log of area of land owned. Sample is restricted to agricultural households with children 6 to 15 years old

Table B.4: Share of farmers under irrigation and with water licenses

Region	% has irrigation	% irrigated farmers has water license
Coast	94.6	53.2
Highlands	49.5	20.0
Total	56.7	25.3

Source: Agricultural Census 2012.